

EEET202

CHINES & TRANSFORMERS

MODULE:2

Parallel Operation of DC Generators

Performance characteristics of DC generators

Prepared By,

ADARSH SR

Asst. Professor

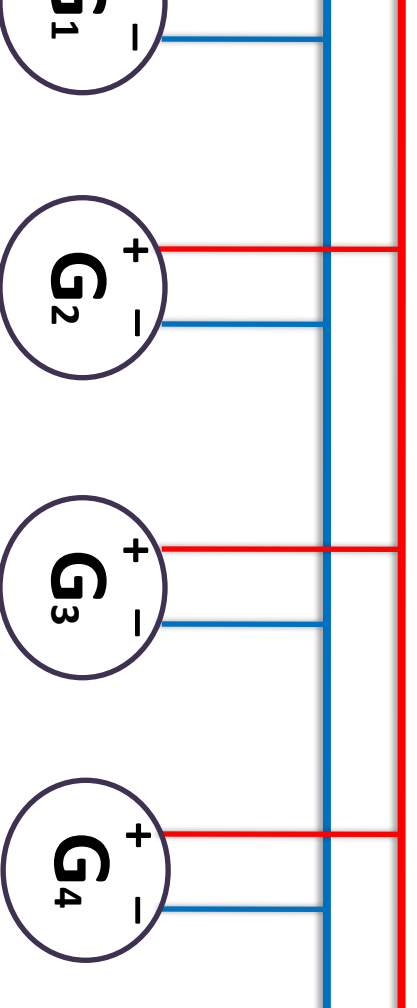
EEE Department

SCET Kodakara



Parallel Operation

Need of Parallel operation
of Service
Maintenance and Repair
Plant Capacity
Availability of Single Large Unit

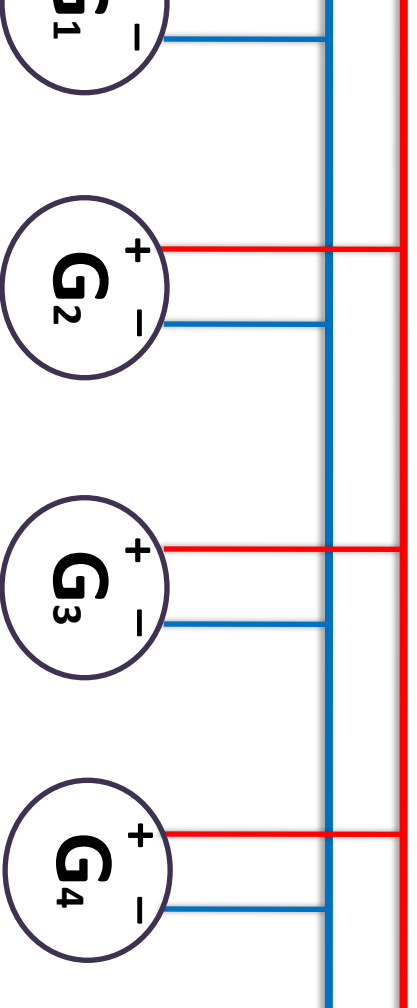


Parallel Operation

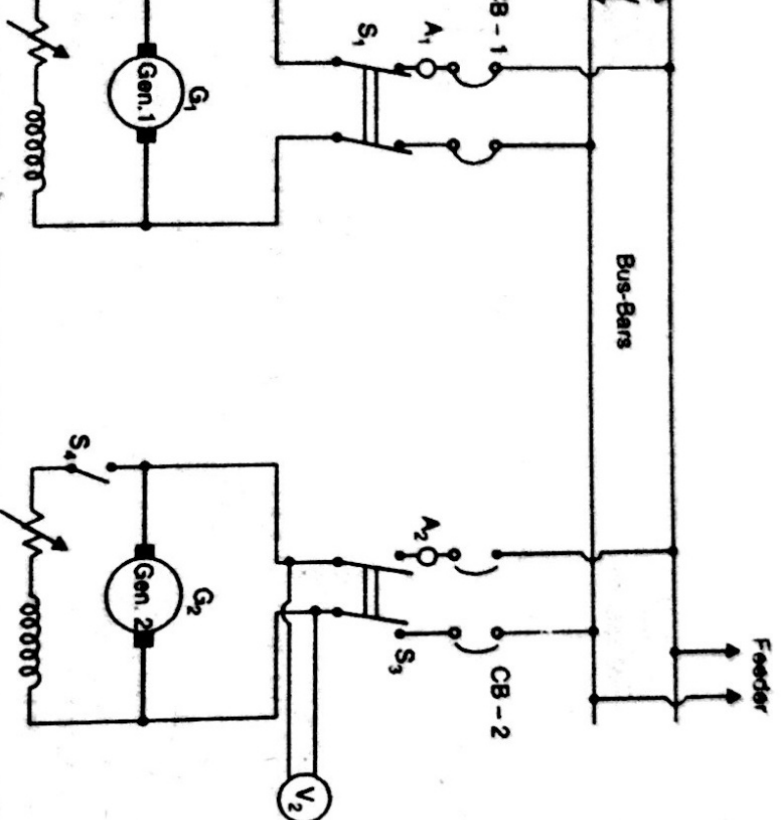
Parallel operation

es must be correct: *+ve terminal must be to +ve bus bar & -ve terminal to -ve bus bar.*

al voltage of generators being paralleled must



Paralleling Procedure

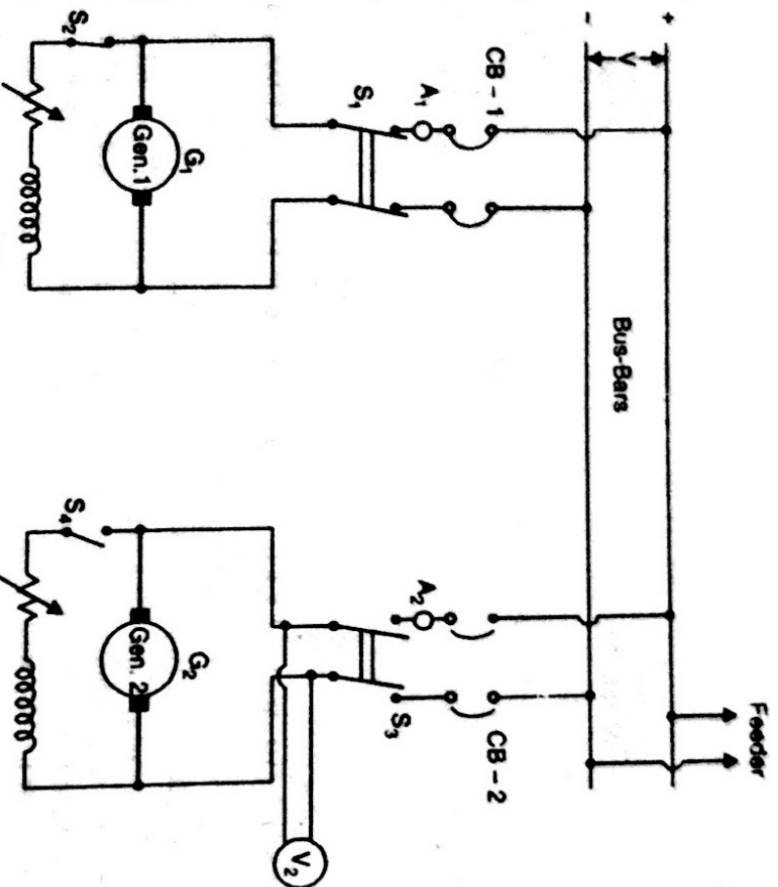


As a shunt generator G_1 connected to the DC bus a load. When the load on the power plant the capacity of G_1 , the second generator G_2 has in parallel with the first, to meet the increased procedure for paralleling G_2 is as follows:

Paralleling Procedure

of G2 is the rated switch S4 in it of G2 is

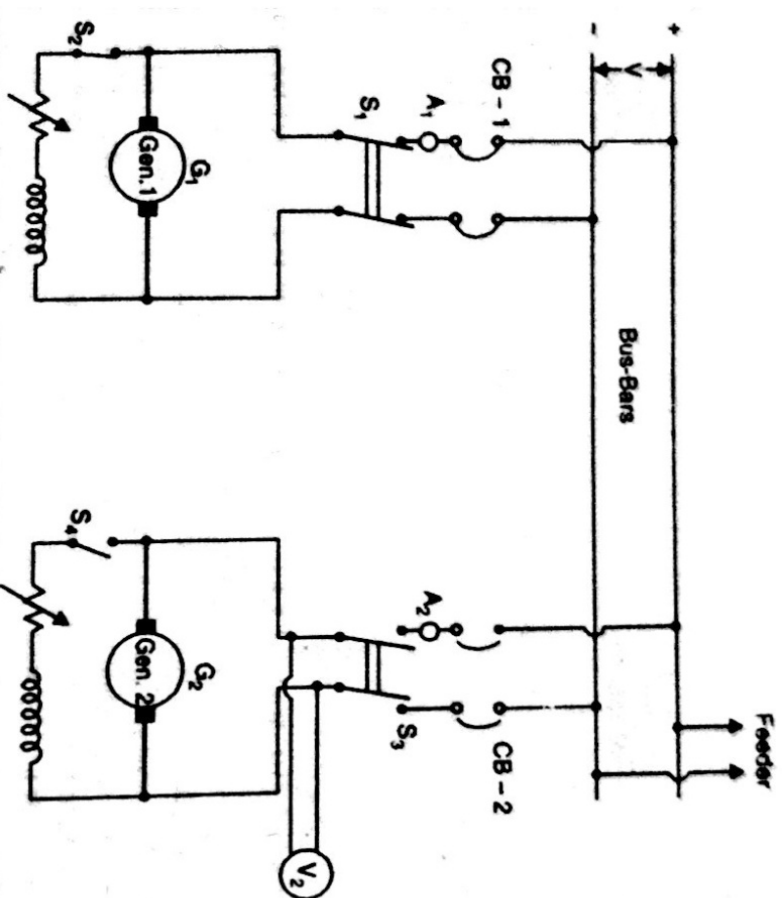
1 of G2 is generates a to bus bar s indicated or V2.



to be paralleled with G1. So close the switch
G2 is not supplying any load because its
age is equal to the bus bar voltage. The
d to be “floating” on the bus bar.

Paralleling Procedure

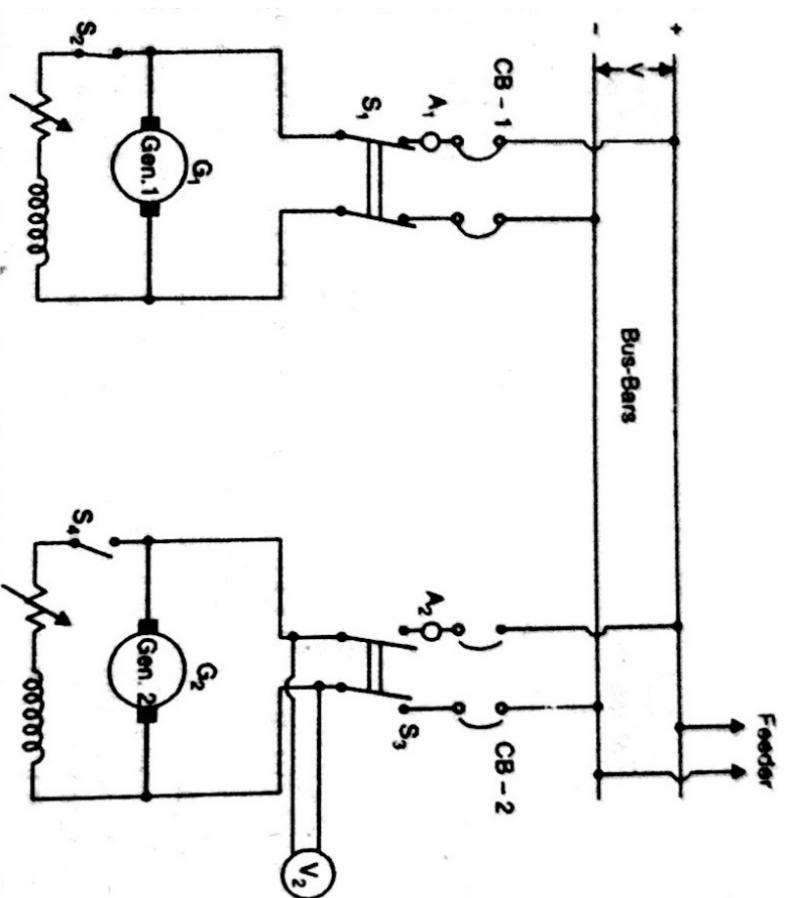
G2 is to
urrent, then
voltage 'E'
reater than
'V'. In that
plied by it



the armature
generator G2

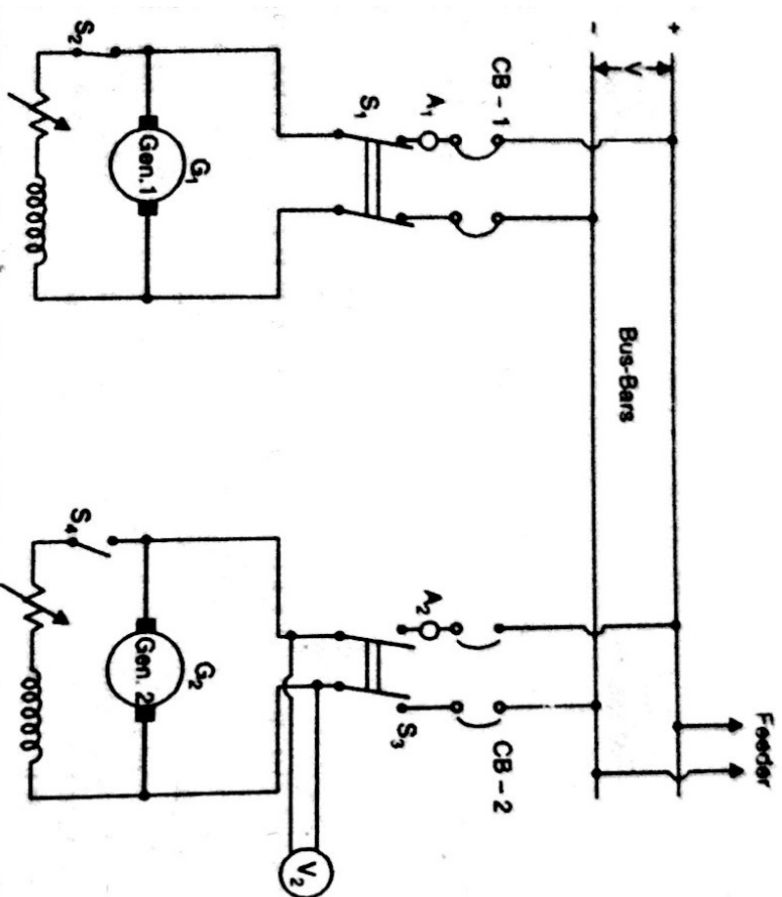
Paralleling Procedure

held current,
e to supply
of load.
be shifted
enerator to
merely
xcitation.
shutdown,
d must be
provided it
y to supply



Paralleling Procedure

reduce the
G1 to make
plied by it to
n the main
hen again
ield current
n S2. At last
e mover of



$$\text{m G1, } I_1 = \frac{E_1 - V}{R_{a1}}$$

$$\text{m G2, } I_2 = \frac{E_2 - V}{R_{a2}}$$

Problems

Two generators, each with armature resistance of $0.1\ \Omega$ and field resistance of $50\ \Omega$ run in parallel and supply a current of 1000 amperes to the load circuit. If the terminal voltages are 270 V and 265 V , what will be their bus bar

Problems

Two generators with induced emfs of 120V and 110V have internal resistances of 0.05 Ω and 0.04 Ω and field resistances of 20 Ω and 25 Ω respectively are in parallel with a total load of 25kW. Calculate the load shared by each generator?